

HEAT TRANSFER LAB

III B. Tech. II Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6ME32	PCC	0	0	2	1	40	60	100

COURSE OVERVIEW:

The Heat Transfer Lab enables the student to understand the concepts of conduction, convection and radiation practically in various thermal systems through a series of experiments. Students have a fairly good understanding of the theory underlying the experiments and the entire lab course is designed such that classroom lectures precede the lab work.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Apply the principles of various modes of heat transfer to evaluate thermal conductivity of different metals and non-metal objects under steady state conditions.
2. Apply the principles of various modes of heat transfer to evaluate thermal conductivity of different objects under unsteady state conditions.
3. Estimate heat transfer coefficients in forced convection, free convection, condensation and correlate with theoretical values.
4. Obtain variation of temperature along the length of the pin fin under forced and free convection.
5. Apply the principles of radiation to evaluate the surface emissivity of a test plate and Stefan-Boltzmann's constant and compare with theoretical values.

LIST OF EXPERIMENTS:

1. Composite Slab Apparatus – Overall heat transfer co-efficient.
2. Heat transfer through lagged pipe.
3. Thermal Conductivity of given metal rod.
4. Heat transfer through pin-fin
5. Heat transfer in forced convection apparatus.
6. Heat transfer in natural convection
7. Parallel and counter flow heat exchanger.
8. Emissivity apparatus.
9. Stefan Boltzmann Apparatus.
10. Heat transfer in drop and film wise condensation.
11. Critical Heat flux apparatus.
12. Study of heat pipe and its demonstration.